

Calibration transfer under limited target labels: a five-benchmark comparison of correction and representation methods

Panlin Li^a, Yuchang Li^a, Longjie Li^{b,c}, Ya Liu^d, Yutong Feng^{b,c}, Muhammad Waqar Akram^e, Junxian Guo^{b,c} and Hua Huang^{a,f,*}

^aCollege of Mathematics and Physics, Xinjiang Agricultural University, Urumqi 830052, China

^bCollege of Mechanical and Electrical Engineering, Xinjiang Agricultural University, Urumqi 830052, China

^cXinjiang Key Laboratory of Intelligent Agricultural Equipment, Urumqi 830052, China

^dUrumqi Comprehensive Experimental Station, Xinjiang Academy of Agricultural Sciences, Urumqi 830013, China

^eDepartment of Agricultural Engineering, University of Agriculture Faisalabad, Faisalabad 38000, Pakistan

^fResearch Center for Numerical Simulation and Data Analysis, Xinjiang Agricultural University, Urumqi 830052, China

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ABSTRACT

Near-infrared calibration models degrade across instruments, seasons and origins, yet when a deep or physics-informed representation beats the simplest correction under controlled label budgets is unevaluated. One protocol spans five benchmarks (corn, tablets, soil, mango, apples), four shift types and two modalities: 126 transfer tasks, five seeds, task-clustered bootstrap, errors in native units and as NRMSEP ($= 1/\text{RPD}$). At 0–40 target labels, label-free second-moment alignment (CORAL, transductive) and a two-parameter slope/bias correction match or exceed the deep and physics-informed baselines each budget admits on the instrument, seasonal and cross-origin benchmarks (on the last, no method including PLS is far from a constant predictor in-domain); cross-laboratory soil in the mid-infrared is the exception, the deep family ahead at every positive budget; a selection-free reanalysis agrees. The winner map fails as a rule (held-out median regret 11.2%), while the fixed prescription “CORAL at zero labels, slope/bias otherwise” needs no shift diagnosis and had the lowest leave-one-benchmark-out regret (2.8% against 20.8% for a default deep model). Shift magnitude does not predict the crossover budget ($R^2 \leq 0.13$); structure partly does: first- and second-moment matching removes 92% of the source–target distance under instrument shift, 73–78% under population shifts, and this share, not the distance, correlates with the CORAL gain ($\rho = 0.53$) and the simple family’s zero-label margin ($\rho = 0.49$). A label-free low-rank index warns of direct-transfer failure ($\rho = 0.49$, just under its pre-registered 0.5). In the apple case study (Supplement), a Beer–Lambert-inspired architecture’s edge over a generic network reverses once the same labels fund slope/bias correction.

1. Introduction

Near-infrared (NIR) and mid-infrared (MIR) spectroscopy combined with chemometric modelling has become a mature technique for rapid, non-destructive quantification in agricultural produce quality, soil properties and pharmaceutical quality control[1–4]. Under the controlled conditions of a single instrument and a single batch, models such as partial least squares regression (PLSR)[5] frequently reach a ratio of performance to deviation (RPD) above 2, the level commonly used as a screening-grade criterion, although the criterion itself has been criticised[6]; apple soluble solids content is a well-studied case[7, 8]. Such calibration models, however, generally lose accuracy when deployed across instruments, batches, seasons, growing origins or laboratories, which is what makes calibration transfer necessary at all[9, 10]. Calibration transfer, bridging the systematic spectral distribution shift between a source domain and a target domain

while using as few target-domain calibration labels as possible, or none at all, has therefore become a bottleneck common to the large-scale deployment of quantitative spectroscopy, spanning grain, tablets, soil and fruit[9–11]. For apple soluble solids content (SSC, i.e. sugar content), our group has previously established the effectiveness of visible/near-infrared spectroscopy combined with functional linear regression and fractional-differential ensemble learning[12, 13]; performance nonetheless degrades substantially under cross-origin deployment; this setting supplies one of the five benchmarks below and a compact case study of a physics-informed architecture.

Around this bottleneck the methodological landscape has expanded into four families of increasing cost. Spectral preprocessing, multiplicative scatter correction, standard normal variate transformation and derivatives, partially normalises scattering differences[14–16]. Empirical calibration transfer, direct standardisation, piecewise direct standardisation (PDS)[17], slope/bias correction, second-order moment alignment (correlation alignment, CORAL)[18] and

*Corresponding author.

✉ huanghua@xjau.edu.cn (H. Huang)

their recent variants[8, 19–22], is effective against inter-instrument differences and batch drift but mostly needs paired or a few target calibration samples[23–26]. Statistical and deep domain adaptation[27] aligns source and target feature distributions with few or no target labels, through domain-adversarial networks (DANN)[28], deep second-order moment alignment (Deep CORAL)[29], regularisation built on the maximum mean discrepancy (MMD)[30] kernel-subspace alignment of calibration models[31, 32] and convolutional networks trained on spectra[33–39]. Physics-informed learning encodes priors such as the Beer–Lambert (BL) law into the network, in the hope of robustness where data are scarce and the deployment extrapolates[40–43]; a contrasting line drops the law and learns the concentration map outright[44].

Methods have grown steadily more complex and more expensive, yet one question of first-order practical importance has never been answered systematically: given a concrete calibration transfer and an affordable target-domain labelling budget, when is it worth deploying an expensive deep or physics-informed representation, and when does the simplest classical correction already suffice? Comparative evaluations of calibration transfer exist[9–11, 45], but they are set in instrument transfer, and we are not aware of one that varies the target-label budget across several types of shift; studies introducing deep or physics-informed transfer are almost always conducted on a single dataset and a single shift type, with simple baselines under-represented. Two further questions follow. In practice the shift is often mixed and its severity unpredictable, so a recommendation that presupposes a shift diagnosis is of limited use; what is needed is a prescription whose cost of being wrong has been measured on data it was not fitted to. And whether a “transferability scaling law” exists, one predicting from measurable shift magnitudes (distribution distances, intrinsic dimensionality) the crossover budget n^* at which a simple method catches up with a complex one, has never been genuinely tested; if it does not hold, the question becomes whether some structural property of the shift does.

Under a unified protocol, this study examines five calibration-transfer benchmarks spanning four types of domain shift and two spectral modalities, corn across three instruments[46], tablets across two instruments (IDRC 2002 shoot-out)[47], global soil mid-infrared across laboratories (Open Soil Spectral Library, OSSL)[48], mango across seasons[49], and in-house apples across origins and years, comprising 126 transfer tasks, and, with five fixed random seeds and bootstrap inference clustered on task, provides a controlled empirical comparison of three families of methods (simple classical corrections, classical model updating, and deep/physics representations) at each label budget. The objectives are: (1) to reckon simple

corrections against deep and physics-informed representations at each budget and report, by shift type, the direction and size of the simple family’s margin; (2) to quantify label-free second-order moment alignment, organise the most frequent winner over shift type and budget into an in-sample frequency map, and test by leave-one-benchmark-out whether that map or any fixed prescription carries over to an unseen benchmark; (3) to test whether n^* is predictable from shift magnitude; and (4) to test, from spectra alone, whether the structure of the shift explains where the simple corrections win. The in-house apple data, besides supplying one benchmark, carry a compact case study (Section 3.5; Supplementary Sections S1–S12) of a physics-informed factorisation architecture that encodes the BL law as an inductive bias (Phys+BL). The contribution is an evaluation and a diagnosis rather than a new transfer algorithm: we do not presuppose that complex methods are better, we report faithfully where simple baselines suffice and where the two families are indistinguishable, and we state the boundaries of what five benchmarks can establish. Fig. 1 summarises the design and how its parts map onto the sections that follow.

2. Materials and methods

2.1. Five calibration-transfer benchmarks and a unified transfer-task matrix

To compare method selection across benchmarks and modalities, we assembled five public or in-house calibration-transfer benchmarks covering four types of domain shift and two spectral modalities (Table 1). (1) Corn[46]: 80 corn samples acquired on three NIR instruments (m5, mp5, mp6), 1100–2498 nm (700 channels), with moisture, oil, protein and starch contents as the four label attributes; differences between instruments constitute instrument-response shift. (2) Tablet[47]: 655 tablets from the IDRC 2002 shoot-out acquired on two transmission NIR instruments, 600–1898 nm (650 channels), with tablet weight, hardness and active-ingredient assay as the three label attributes; also instrument shift. (3) Soil[48]: the mid-infrared (MIR) subset of the OSSL global soil spectral library, taking the KSSL, AFSIS1 and CAF sub-libraries established by different laboratories; the raw high-dimensional absorbance spectra were binned at equal wavenumber intervals to 340 dimensions, with total carbon and sand content as labels; differences between sub-libraries constitute cross-laboratory shift, the reuse of a large MIR soil library across instruments and settings being a recognised obstacle in its own right[50, 51]; this is also the only MIR modality in this study. (4) Mango[49]: the visible–near-infrared spectra of intact mango fruit from Anderson et al. across four harvest seasons (binned to 103 dimensions), with dry matter (DM) content as the label; differences between seasons constitute

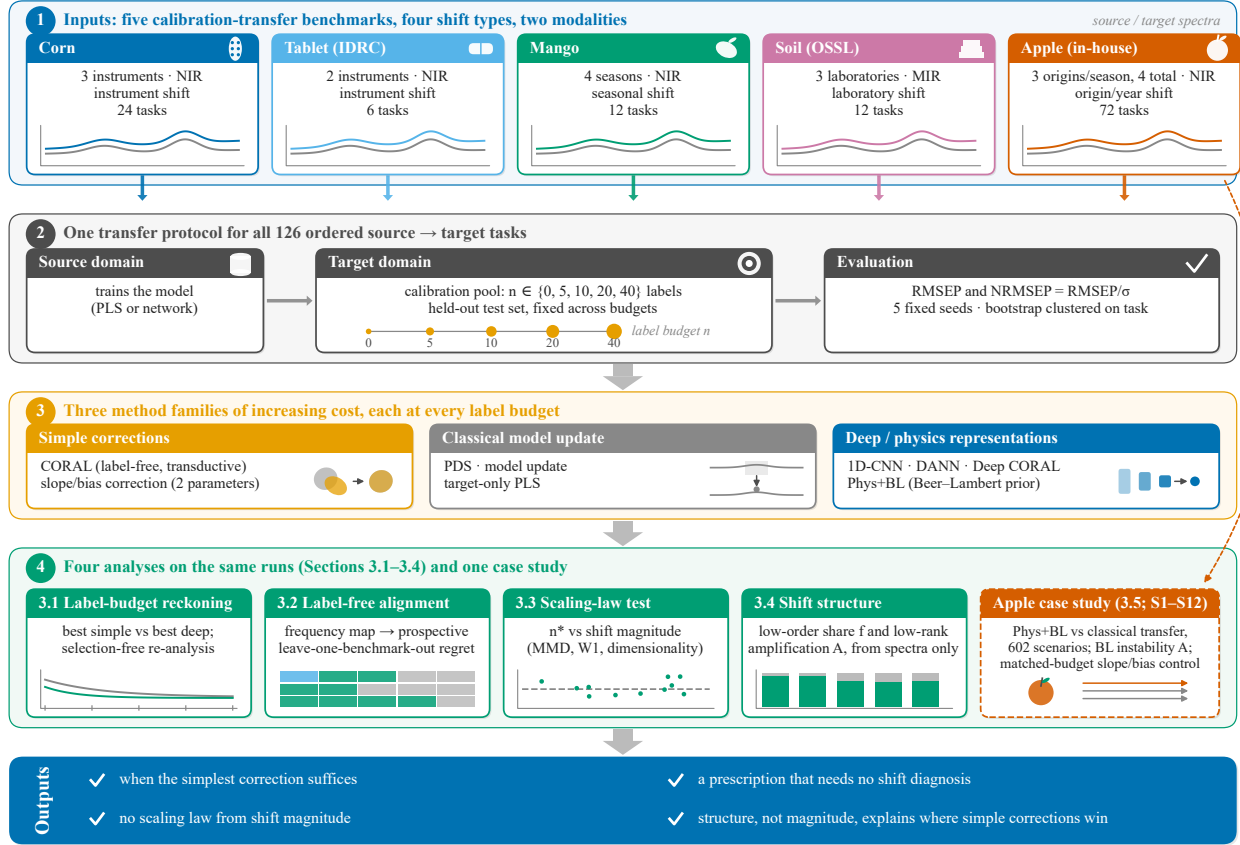


Figure 1: Overview of the study: five calibration-transfer benchmarks (four shift types, two modalities) enter one evaluation protocol; three method families of increasing cost are compared at each target-label budget; four analyses answer, in turn, when the simplest correction suffices, what to do when the shift type is unknown, whether a scaling law exists, and whether the structure of the shift explains the outcome. The in-house apple data also support a compact case study of the physics-informed architecture, reported in the Supplement. Curves and tiles drawn inside the cards are schematic sketches, not data

seasonal shift. (5) Apple (in-house): diffuse-reflectance spectra of intact ‘Fuji’ apples acquired in 2018, 2019 and 2025 across three origins per season and two instrument generations (aligned to 229 common wavelength channels), with SSC as the label. Four origins are involved in total (Shandong, Xinjiang, Shaanxi and Gansu) and the set is not fixed across seasons: Gansu replaced Shaanxi in the 2025 season, so the design is incompletely crossed and any domain pair involving Shaanxi or Gansu carries a difference of origin and of year together. Three seasons with three origins each give nine atomic origin–year domains, whose ordered pairs across all attributes yield the 72 apple transfer tasks; the 2019 season (instrument A, acquired and cleaned as for 2018) enters the unified benchmark but not the case study (Section 3.5), which contrasts the 2018 and 2025 seasons. The compound differences of origin $\times$ year $\times$ instrument constitute cross-origin/year shift. The apple data are summarised in Section 2.4 and detailed in Supplementary Section S1.

Instrument, seasonal, cross-laboratory and cross-origin/year shift are operational categories named after the experimental factor, and none of them corresponds to one pure, independent statistical mechanism; in the dataset-shift taxonomy of the machine-learning literature[52, 53] they fall into different formal classes, and the class matters because it says what a given correction can remove in principle. Instrument shift alters the spectra but not the samples: the marginal $p(\mathbf{x})$ changes while the analyte content and the physical relation between content and absorbance do not, so it is a covariate shift, and one that is close to a low-order transformation of $\mathbf{x}$ (Section 3.4), precisely the structure that standardisation and moment-alignment methods target. Seasonal, cross-origin and cross-laboratory shift change the sample population as well: the distribution of the analyte $p(y)$ moves (prior or target shift), and because matrix composition, tissue structure, moisture and particle size co-vary with season, origin and laboratory, the conditional $p(\mathbf{x} | y)$ changes too (conditional shift[54]); the cross-laboratory

soil case in addition combines a genuine instrument component with a population component, and the cross-year apple pairs add an instrument change to the population change. Under conditional shift the invariance assumption behind feature alignment (matching the marginals of $\mathbf{x}$) no longer guarantees correct predictions, and some target labels become necessary in principle. Table 1 records this classification for each benchmark; we return to it when interpreting where label-free alignment stops working (Sections 3.2 and 3.4).

Transfer tasks were formed by combining the ordered domain pairs within each benchmark with the label attributes: the source domain provides the calibration and the target domain is the condition to be transferred to. This yields 126 transfer tasks, corn 24 (6 ordered instrument pairs $\times$ 4 attributes), tablet 6 (2×3), mango 12 (4 seasons pairwise ordered $\times$ 1 attribute), soil 12 (6 ordered laboratory pairs $\times$ 2 attributes) and apple 72 (ordered multi-domain pairs $\times$ SSC), which by shift type comprise 30 instrument, 72 cross-origin/year, 12 seasonal and 12 cross-laboratory tasks, and by modality 114 NIR and 12 MIR tasks. All benchmarks share the same evaluation pipeline: the full source domain trains the source model; the target domain is randomly partitioned into a held-out test set and a disjoint calibration pool, from which the n target calibration labels of each budget $n \in \{0, 5, 10, 20, 40\}$ are drawn. The same held-out test samples serve every budget and every method within a given draw, so differences along a budget curve reflect adaptation rather than a changing evaluation set, and the $n = 0$ methods (direct transfer and label-free CORAL) are scored on exactly those test samples, and the partition is redrawn and the task repeated independently under five fixed random seeds; the exact definition of a target label and the use of unlabelled target spectra are given in Section 2.3. The source, licence and SHA-256 checksum of every dataset are given in the data card (Table 2); the public benchmarks are not re-hosted here and should be obtained from their official sources.

2.2. Three families of methods

Three families of methods, of increasing complexity and cost, were compared under the same pipeline. Simple classical corrections: (i) second-order moment alignment, CORAL[18], which uses only unlabelled target spectra to align the covariance of source-domain features to that of the target domain and uses no target labels whatsoever; (ii) slope/bias correction (SBC), which corrects the scale and offset of the source model's predictions by univariate linear regression on a small number of target-domain calibration samples, using only two parameters. Classical model updating (requiring more target labels): piecewise direct standardisation, PDS[17], which stands for the paired-standard family because its members regularise the

same slave-to-master mapping rather than consume different information (direct standardisation, the full-window limit of that mapping, was run on the same tasks as a check and is reported in Supplementary Section S13); model updating (retraining after pooling source-domain and target-domain calibration samples); and target-only calibration (building a PLS model from scratch on target-domain calibration samples alone, without using the source domain). Deep/physics representations: a one-dimensional convolutional neural network (CNN); the domain-adversarial network DANN[28]; deep second-order moment alignment, Deep CORAL[29]; and the physics-informed BL factorisation architecture Phys+BL (Supplementary Section S4). For fairness, all methods share the same source/target split and random seed within each task and operate on standard-normal-variate (SNV) standardised spectra, so that zero-label direct transfer of the source PLS model is itself a PLSR+SNV baseline. The input, the target-label requirement, the parameter scale and the applicability of every method at each budget are summarised in Table 3: at zero labels only six label-free methods can run, and methods that need target labels cannot be fitted at that budget, so the $n = 0$ comparison is one among label-free methods rather than a fair contest of all methods under the same conditions; a method that cannot run is recorded as not applicable and is excluded from the win-frequency count at that budget. Network architecture, epochs, learning rates and early-stopping rules are given in Supplementary Section S13. Note that this unified benchmark uses SNV standardisation, whereas the apple case study (Section 3.5) is a separate pipeline with its own preprocessing (wavelet denoising and Savitzky–Golay smoothing, scatter correction deliberately omitted, Supplementary Section S2), so absolute numbers are not comparable between the two.

2.3. Label-budget reckoning, the crossover budget and the statistical protocol

The budget n counts target-domain samples that a method must obtain before it can be fitted. For every method but one that sample is a “target label”: a target-domain sample whose reference value the method uses for fitting. The exception is PDS, which uses no reference value but does require n target-domain samples re-measured on the second instrument as paired transfer standards; we charge it the same n because the sample must be obtained either way, and note that under an accounting that counted reference measurements alone PDS would be a label-free method restricted to the paired-instrument benchmarks and would enter the $n = 0$ comparison (Section 4.4). With that convention, slope/bias correction, model update, target-only calibration and the fine-tuning of the deep models use exactly these n samples and reserve none for validation; every deep source model (the CNN,

Table 1

Composition of the five calibration-transfer benchmarks and the transfer-task matrix

Benchmark	Shift type	Dataset-shift class	Modality	Domains (n)	Bands	Label attributes	Tasks
Corn	Instrument	Covariate	NIR	3 instruments	700	moisture, oil, protein, starch	24
Tablet	Instrument	Covariate	NIR	2 instruments	650	weight, hardness, assay	6
Mango	Season	Prior + conditional	NIR	4 harvest seasons	103	dry matter	12
Soil (OSSL)	Laboratory	Covariate + conditional	MIR	3 sub-libraries	340	total carbon, sand	12
Apple	Origin/year	Prior + conditional (+ covariate)	NIR	origin-year-instrument	229	SSC	72
Total	4 types		2 modalities				126

Note: “Domains (n)” is the number of atomic acquisition conditions within the benchmark; “Tasks” is the number of combinations of ordered domain pairs with label attributes. “Dataset-shift class” follows the taxonomy of Section 2.1: covariate shift changes $p(\mathbf{x})$ only, prior shift changes $p(\mathbf{y})$, conditional shift changes $p(\mathbf{x} | \mathbf{y})$; the cross-year apple pairs add an instrument (covariate) component, and soil combines laboratory instrumentation with different soil populations. Corn, tablet and soil are public benchmarks; mango is the public dataset of Anderson et al.; apple was acquired in house (Supplementary Section S1). Soil is the only mid-infrared (MIR) modality.

Table 2

Data card: source, licence and traceability checksum for every dataset used in this study. The upper block lists the four public benchmarks (obtained from their official sources and not re-hosted here); the lower block lists the three in-house apple year-domains that make up the cross-origin benchmark.

Dataset	Source	Licence / availability	SHA-256 prefix (16 hex)
Corn (Cargill)[46]	Eigenvector Research	Free for research use ^a	e28fd4be274a54ca
Tablet (IDRC 2002 shoot-out)[47]	Eigenvector Research	Free for research use ^a	129a32ec9e194e56
Soil (OSSL)[48]	OpenGeoHub / Soil Spectroscopy	CC-BY 4.0	2044946ad5a10af0
Mango (Anderson)[49]	Mendeley Data	CC-BY 4.0	86159e2df9244fa5
Apple 2018, instrument A (in-house)	Xinjiang Agric. Univ.	Restricted ^b	a248445fd874bbf0
Apple 2019, instrument A (in-house)	Xinjiang Agric. Univ.	Restricted ^b	1495630ea432f4fb
Apple 2025, instrument B (in-house)	Xinjiang Agric. Univ.	Restricted ^b	bc6337e462fe5006

Note: the SHA-256 column is the first 16 hexadecimal characters of the SHA-256 checksum, of the raw source file for the public benchmarks, and of the cleaned per-year table (produced by the preprocessing script) for the apple data, and matches the values registered in the code repository’s data cards. ^a Free for research use per the publisher’s terms. ^b Internal data of Xinjiang Agricultural University, not publicly released; available from the corresponding author (huanghua@xjau.edu.cn) on reasonable request, subject to the university’s data-sharing policy. Public benchmarks are not re-hosted; please download them from the official sources cited above and cite the original works.

Phys+BL, DANN and Deep CORAL) is trained under the same fixed budget of optimisation steps and early-stopped on a source-domain hold-out of about one sixth of the source samples that enters no loss term (13 spectra on corn, the smallest benchmark); fine-tuning runs for a fixed number of epochs; so early stopping and weight selection touch no target label at any budget. Each draw splits the target domain at random into two halves, a test set that is fixed for every budget and every method within that draw and a calibration pool from which the n calibration samples of each budget are drawn independently (budgets are not nested). The classical methods repeat this draw five times per fixed seed (25 runs per cell), the deep methods once per seed (5 runs per cell), and the medians below are taken over all runs. The two families are therefore estimated with unequal precision. The two analyses below bracket what that asymmetry can do: the best-of-family envelope of Table 4 takes a per-task minimum and so favours the noisier side, whereas the family mean of Table 5 averages and so penalises

it. The use of unlabelled target spectra is stated per method in Table 3: CORAL estimates its second-order moments transductively from all target spectra (the spectra, not the labels, of the test samples included), DANN and Deep CORAL use only the unlabelled spectra of the calibration pool, PDS uses the spectra of n paired transfer standards on both instruments (their labels are not needed), and no other method uses any unlabelled target spectrum; test samples therefore never enter the fitting of any method, the single exception being CORAL’s transductive covariance estimate, which is why zero-label usability is not inductive deployment (Section 4.4). One property of the paired-instrument benchmarks should be stated explicitly: corn and tablets are the same physical specimens measured on two or three instruments, so the source model is fitted on the source-instrument spectra and reference values of specimens whose target-instrument spectra later form the test set. This is inherent to instrument-transfer data and applies equally to every method compared here, but it means that on those two

Table 3

Method applicability matrix: input, parameter scale, applicability at each target-label budget, and role in the comparison. The zero-label comparison is not a fair contest of all methods under the same conditions: methods that need target labels cannot be fitted at that budget, so at $n = 0$ only label-free methods are compared.

Method	Input	Parameters	$n = 0$	$n = 5-40$	Role in the comparison
Direct transfer (source PLS)	source calibration	PLS	✓	✓ ^a	zero-label floor
CORAL	source calibration + all unlabelled target spectra (transductive)	closed form, 0	✓	✓ ^a	label-free alignment
SBC	source-model predictions + n target labels	2	—	✓	cheapest labelled correction
PDS	spectra of n paired transfer standards on both instruments	per-channel ridge	—	✓ ^b	classical standardisation, paired-instrument benchmarks only
Model update	source calibration + n target labels	PLS	—	✓	simplest labelled update
Target-only	n target labels only	PLS	—	✓	lower-bound sentinel: is the source domain useful
CNN	source labels; fine-tuned on n target labels when $n > 0$	10^3-10^4	✓(zero-shot)	✓(fine-tuned)	plain deep baseline
DANN	source labels + unlabelled spectra of the calibration pool	10^3-10^4	✓	✓ ^a	deep domain adaptation
Deep CORAL	source labels + unlabelled spectra of the calibration pool	10^3-10^4	✓	✓ ^a	deep counterpart of CORAL
Phys+BL	source labels; fine-tuned on n target labels when $n > 0$	10^3-10^4	✓(zero-shot)	✓(fine-tuned)	physics-informed member

Note: ^a label-free methods consume no target label, so their result at $n > 0$ is their $n = 0$ result (same test samples); they enter the oracle candidate set of Table 6 at $n = 0$ but neither the $n > 0$ cells of the frequency map of Fig. 3a nor the $n > 0$ rows of Tables 4 and 5, which compare the two families among the methods that consume the label budget. ^b Feasible only on corn and tablets, where the same physical samples were measured on both instruments. Hence the denominator of the win frequencies in Fig. 3a is the 6 label-free candidates at $n = 0$ and, at $n \geq 5$, the 6 label-consuming candidates on the instrument benchmarks and 5 elsewhere (of which the classical members are 2, 4 and 3); twelve candidates are run in each instrument benchmark and eleven elsewhere, but the oracle of Table 6 is taken at each budget over the methods that consume it: the 6 label-free candidates at $n = 0$, and at $n \geq 5$ the 6 that take target labels on the instrument benchmarks and 5 elsewhere. Keeping the label-free candidates in the candidate set at every budget is reported as a sensitivity in Section 3.2. The deep methods share one convolutional backbone (Supplementary Section S13).

benchmarks the comparison is one of spectral transfer, not of generalisation to unseen specimens.

For each (task, method, label budget) cell, the median root mean square error over all runs (the five fixed seeds and their repeated draws; denoted RMSEP) was taken as the performance of that cell, reported in the native units of the target variable. Alongside it, as a co-primary endpoint, we report the normalised error $\text{NRMSEP} = \text{RMSEP}/\sigma_{y,\text{tgt}} = 1/\text{RPD}$, which is dimensionless and comparable across analytes and benchmarks. Every pooled or cross-benchmark result below is reported on both endpoints: Tables 4 and 5 carry the median error of each family on both scales and the dominance statistic on the original scale, with its NRMSEP-scale counterpart reported in Section 3.1, since a rank statistic pooled over tasks is not invariant to the per-task rescaling that NRMSEP applies. Usability was expressed as $\text{RPD} = \sigma_{y,\text{tgt}}/\text{RMSEP}$ (following common but application-dependent screening conventions[6], $\text{RPD} \geq 2$ is treated as screening-level and ≥ 1.4 as preliminary). The core comparison is the task-wise paired difference between the best simple method (the better of CORAL and SBC) and the best deep/physics method (the best of CNN/DANN/Deep CORAL/Phys+BL) on each task: significance by the

Wilcoxon signed-rank test (the distributions being heavy-tailed and non-normal), effect size reported as Cliff's δ , confidence intervals from a cluster bootstrap clustered on task (≥ 2000 resamples), and multiple comparisons corrected by the Holm procedure within a pre-specified family of hypotheses. Cliff's δ is the unpaired dominance statistic, the fraction of all cross-family task pairs on which the simple family has the smaller error minus the fraction on which the deep family does, positive values favouring the simple family, and we report alongside it the paired per-task win fraction ("Simple-win %" in Tables 4 and 5), so that a δ of +1 is transparently the case of complete separation between the two error distributions and the win fraction its paired complement. Because taking the per-task best within each family is a selection that favours the larger (deep) family, the pre-specified selection-free reanalysis (Table 5) rather than the best-of-family envelope is treated as the primary basis for the simple-versus-deep comparison.

The crossover budget n^* is defined, consistently throughout, as the minimum target-label budget on the grid $\{0, 5, 10, 20, 40\}$ at which the best simple method catches up with or overtakes the best deep/physics method; we additionally define n_{usable} as the minimum

number of labels required to reach a given RPD threshold. Because n^* is read off a discrete budget grid it is interval-censored (and right-censored when no crossover occurs by 40 labels), so the log–log fits below are a deliberately conservative test of predictability rather than an exact continuous regression. To test the “transferability scaling law”, we regressed n^* on measures of domain shift in log–log space; the predictors comprise the maximum mean discrepancy (MMD) and the sliced Wasserstein distance between source and target distributions, the intrinsic dimensionality of the target domain (effective PCA dimension) and the signal-to-noise ratio. Extrapolability was tested by leave-one-benchmark-out (LOBO: fit on the remaining benchmarks, predict the held-out benchmark) together with conformal 90% prediction intervals. All formal experiments used the five fixed random seeds [20060515, 20041210, 19810915, 2023, 2024].

Because the frequency map of Section 3.2 is fitted in-sample, we test by leave-one-benchmark-out whether it, or any fixed prescription, carries over to a benchmark it has not seen. For each (task, budget) the oracle is the method with the smallest across-seed median RMSEP among all twelve candidates run in the benchmark (direct transfer, CORAL, SBC, PDS, model update and target-only calibration; the zero-shot and the fine-tuned variant of the CNN and of Phys+BL; and DANN and Deep CORAL, which consume no target label and therefore have a single variant each), eleven where PDS is infeasible. The candidate set of a budget is the set of methods that consume that budget: the six label-free candidates at $n = 0$, and the six that take target labels at $n \geq 5$ (five where PDS is infeasible), the same convention as in Tables 4 and 5, so that each row asks what the labels buy rather than comparing against a label-free result carried forward. Keeping the label-free candidates at every budget instead is reported as a sensitivity in Section 3.2. The regret of a prescription is $\text{RMSEP}_{\text{chosen}}/\text{RMSEP}_{\text{oracle}} - 1$, a dimensionless quantity on the same test samples that can be pooled across benchmarks. Two kinds of prescription are scored by leave-one-benchmark-out. (i) The frequency map used as a rule: for each held-out benchmark the map is re-derived from the other four alone with each task weighted equally and its most frequent winner is applied, either with the held-out benchmark’s shift type known (possible only for the instrument type, which two benchmarks share) or unknown (the most frequent winner over all training tasks, each task counted once). (ii) Fixed prescriptions that need no knowledge of the shift: the simplest applicable correction (CORAL at zero labels, SBC otherwise), always model update, always target-only calibration, and a default deep model (zero-shot CNN at zero labels, fine-tuned CNN otherwise). Medians and 95% intervals come from a bootstrap clustered on task (2000 resamples), and the fraction of task–budget cells

within 10% of the oracle is reported alongside (script 73). Finally, because a deep model that fails in the target domain and a deep model that never trained are indistinguishable from the target-domain error alone, the four deep baselines were also scored on held-out samples of their own source domains, against a PLS regression and a constant predictor on identical splits; Supplementary Section S14 reports that control.

2.4. Apple benchmark and the label-free diagnostics of shift structure

The in-house apple benchmark is described here only as far as is needed to read the unified results; acquisition, cleaning and the case-study design are given in Supplementary Sections S1–S7. Intact ‘Fuji’ apples at commercial maturity were scanned in diffuse reflectance over 591–1001 nm in three seasons (2018 and 2019 on one spectrometer, instrument A, and 2025 on a second-generation instrument from the same manufacturer, instrument B, aligned to 229 common channels) and from three growing origins per season (2018 and 2019: Xinjiang, Shandong, Shaanxi; 2025: Xinjiang, Shandong, Gansu), with reference SSC by digital refractometry. Three cleaning rules are applied uniformly to these data and shared by every experiment in this paper, the unified benchmark and the case study alike: at preprocessing, a fixed-quantile spectral outlier removal (10%; the 0% and 15% alternatives leave the method ranking unchanged), and, at the single data-reading entry point, removal of physiologically impossible reference values (SSC above 20 °Brix) and removal of bit-wise identical spectra attributable to an acquisition defect, which leaves 1679 fruits in nine origin–year domains (Supplementary Section S1). In the unified benchmark the apple spectra are treated exactly like the four public benchmarks (SNV, Table 3); the finer case study uses its own preprocessing and is summarised in Section 3.5.

Phys+BL, the physics-informed member of the deep family, reconstructs each input spectrum as a non-negative low-rank product of K generalised component spectra, a scalar effective path length and generalised concentrations, the additive skeleton of the Beer–Lambert law, relaxed to a structural prior in latent space, and adds the reconstruction error to the regression loss (Supplementary Section S4). Because the inputs are standardised reflectance rather than absorbance, the factorisation is Beer–Lambert-inspired rather than a physical decomposition, and no latent component is assigned to a constituent (Supplementary Section S2). In the unified benchmark it shares the convolutional backbone of the other deep methods (Supplementary Section S13) with $K = 8$; the case study uses a fully connected $K = 3$ variant.

Two label-free diagnostics of the structure of a shift are computed for every ordered source–target domain

pair (98 pairs) from spectra alone, on the same nearest-wavelength-aligned, SNV-standardised inputs as the benchmark (script 74). (i) Low-order share of the shift. Let $d(X_s, X_t)$ be a distance between the source and target spectral distributions; we use both the squared maximum mean discrepancy with a Gaussian kernel and the sliced Wasserstein-1 distance, with the kernel bandwidth (median heuristic) and the 100 random unit projections fixed once per pair so that the three states below are compared on identical footing. The target spectra are translated to the source mean (X_t^{mean}), and separately matched in first and second moments by the same CORAL implementation as in the benchmark (X_t^{CORAL}), and

$$f_{\text{mean}} = 1 - \frac{d(X_s, X_t^{\text{mean}})}{d(X_s, X_t)} \quad (1)$$

$$f_{\text{CORAL}} = 1 - \frac{d(X_s, X_t^{\text{CORAL}})}{d(X_s, X_t)} \quad (2)$$

are the fractions of the shift removable by matching the first, and the first two, moments (Eqs. (1) and (2)); $1 - f_{\text{CORAL}}$ is the higher-order residual that no first- and second-moment correction can remove. (ii) Low-rank residual amplification. A ten-component principal-component basis is fitted on the mean-centred source spectra; for any spectrum $\mathbf{x}$ the relative reconstruction residual is $\Delta R(\mathbf{x}) = \|\mathbf{x} - P(\mathbf{x})\|_2 / \|\mathbf{x}\|_2$, and

$$A = \frac{\text{median}\{\Delta R(\mathbf{x}) : \mathbf{x} \in X_t\}}{\text{median}\{\Delta R(\mathbf{x}) : \mathbf{x} \in X_s\}} \quad (3)$$

measures how much further the target spectra depart from the source domain's additive low-rank structure than the source spectra themselves ($A > 1$, Eq. (3)). This is the indicator introduced for the apple case study (Supplementary Section S4), now applied to all five benchmarks; a held-out variant (basis fitted on 70% of the source spectra, residuals evaluated on the remaining 30% and on the target) guards against in-sample optimism. Domains larger than 800 spectra are subsampled once with a fixed seed, and the kernel distance uses 300 spectra per domain. Each diagnostic is compared across the 126 tasks with three outcomes taken from the benchmark runs: the zero-label NRMSEP of direct transfer, the relative error reduction achieved by CORAL, $1 - \text{NRMSEP}_{\text{CORAL}} / \text{NRMSEP}_{\text{direct}}$, and the family-mean deep-minus-simple NRMSEP margin at 0 and 20 labels. Associations are Spearman rank correlations with 95% intervals from a bootstrap clustered on domain pair (2000 resamples); their portability is checked by leave-one-benchmark-out and within-benchmark correlations.

3. Results

3.1. Label-budget reckoning between simple corrections and deep/physics representations

Under the unified protocol, the three families of methods were compared on the 126 transfer tasks of the five calibration-transfer benchmarks in terms of target-domain root mean square error at each target-label budget (median across the five fixed seeds, with the task as the clustering unit). A first, descriptive comparison takes the task-wise paired difference between the best simple method (label-free second-order moment alignment CORAL at zero labels, the two-parameter slope/bias correction SBC once labels are available) and the best deep/physics method (the zero-shot one-dimensional convolutional network, domain-adversarial DANN, DeepCORAL or the physics-informed Beer-Lambert factorisation Phys+BL at zero labels, the fine-tuned CNN or Phys+BL above), tested by the Wilcoxon signed-rank test with effect size reported as Cliff's δ (Table 4). Because this takes the per-task best within each family, it is an oracle upper envelope for both families rather than an operational selection rule, and the families are of unequal size (two simple, four deep), so the envelope should be read as descriptive; a pre-specified, selection-free reanalysis that confirms the pattern is reported below (Table 5).

The results separate sharply by the type of domain shift, and one benchmark runs against the rest. For instrument-response shift (corn across three instruments and tablets across two, 30 tasks), the simple corrections are consistently better than the deep representations at every label budget: median root mean square errors of 0.24–0.30 against 0.32–0.37 for the deep methods, with the simple methods better on 73%–93% of tasks, δ between 0.23 and 0.35 and Wilcoxon $P \leq 0.003$. For seasonal shift (mango across four seasons, 12 tasks), the simple methods are better at every budget on 75%–92% of tasks (δ 0.51–0.83, $P \leq 0.05$). For cross-origin/year shift (apple, 72 tasks), the simple methods are ahead at every positive budget (δ 0.08–0.28 at 5–40 labels, $P \leq 0.010$), while at zero labels the two envelopes are not separated ($\delta = +0.07$, $P = 0.08$). For cross-laboratory mid-infrared soil (12 tasks) the direction reverses: the deep family is better on all 12 tasks at every positive budget ($\delta = -0.26$ to -0.35 , $P = 5 \times 10^{-4}$), its NRMSEP falling from 0.71 at zero labels to 0.48 at 40 while the simple family stays between 1.02 and 0.87; only at zero labels are the two not separated ($P = 0.20$). This is the single benchmark on which a learned representation earns its keep, and Section 3.4 shows it is also the benchmark on which a label-free second-moment alignment recovers the least of the zero-label error, leaving it at the level of predicting the target mean.

The physics-informed Beer–Lambert architecture (Phys+BL), as one member of the deep family, did not beat the simple corrections consistently under any type of shift, including the soil benchmark where the deep family as a whole wins, which agrees with the finer-grained apple case study (Section 3.5). Fig. 2 gives the complete reckoning curves of the three families against budget under the four shifts.

To check that these patterns are not an artefact of taking the per-task best within each family (a winner’s curse, compounded by the four-versus-two family-size imbalance), we recomputed the comparison on the same runs in two selection-free ways (Table 5): (i) a pre-specified representative pair at zero labels, CORAL against its deep counterpart DeepCORAL, both being second-order moment alignment and therefore the closest simple-versus-deep analogue; and (ii) the family mean, averaging over all methods in each family per task rather than taking the best. Neither uses post-hoc selection. Under both, the pattern is preserved, and so is the soil exception. Simple methods retain a large advantage under instrument shift (family-mean δ +0.46 and +0.41 at 0 and 20 labels, simple better on 100% and 90% of tasks) and a clear one under seasonal shift (family mean +0.68 at 20 labels, representative pair +0.57 at zero labels, both in Table 5). Under cross-origin shift the selection-free analyses put the simple family ahead at every budget, including zero (representative pair δ = +0.24, simple better on 74% of tasks, $P < 10^{-5}$; family mean +0.24, 75%, $P < 10^{-6}$), so the marginal deep edge that the best-of-family envelope shows at zero labels under cross-origin shift is a selection effect over the larger deep family and does not survive a selection-free comparison (Supplementary Section S14). On cross-laboratory soil both analyses reproduce the reversal: the family mean gives δ = −0.25 to −0.32 at 5–40 labels with the deep family better on all 12 tasks ($P = 5 \times 10^{-4}$), while at zero labels the two are not separated (δ = −0.08, $P = 0.47$). Nor is the choice of DeepCORAL as the comparator responsible: on the same zero-label runs, CORAL individually outperforms each of the four deep methods (CNN, DANN, DeepCORAL and Phys+BL), on 93–100% of tasks under instrument shift (paired δ +0.87 to +1.00, all $P < 10^{-6}$) and on 72–78% of tasks pooled across the 126 tasks (all $P < 10^{-6}$), so the representative pair is representative rather than selected.

Two further checks address incommensurate units and pseudo-replication when pooling across benchmarks (script 68). On the dimensionless NRMSEP reported alongside RMSEP in Tables 4 and 5 the pattern is unchanged in direction and wider in size, because normalisation removes the between-task scale differences that dilute an unpaired comparison on raw units (family-mean δ +0.80 to +0.89 for instrument, +0.64 to +0.94 for season, +0.28 for origin/year at

zero labels rising to +0.83 at 10 labels, and −0.19 to −0.94 for soil, that is against the simple family; script 86). Taking the benchmark itself as the unit, the most conservative choice with only $n = 5$ units, and averaging the family-mean difference over the tasks of each benchmark, four of the five favour the simple family at zero labels and three of the five at 20, and a benchmark-balanced bootstrap gives a mean NRMSEP difference (deep minus simple) of +0.25 (95% CI [0.01, 0.44]) at zero labels and +0.08 ([−0.15, 0.28]) at 20 labels: the pooled advantage of the simple family is therefore established at zero labels but not at 20, where soil (−0.34) and tablets (−0.03) pull the balanced mean towards zero. A third check replaces the task by the source–target domain pair as the unit of analysis, since the 126 tasks fall on 98 pairs and only 21 target domains: the direction is unchanged everywhere and the effect sizes are at least as large, but on the instrument benchmarks, which collapse to eight pairs, the advantage above zero labels is no longer significant (script 101; ranges in the note to Table 5). The direction of the reading is not an artefact of the error unit, of within-benchmark replication or of the choice of clustering unit, but the benchmark-level test is necessarily low-powered and does not support a claim of a pooled simple-family advantage once labels are available.

3.2. Label-free alignment and the in-sample frequency map

We next separate zero-label feasibility from the budget dependence of the best method. Label-free second-order moment alignment (CORAL) recovers accuracy substantially without using any target labels, and the extent of the recovery depends on the type of shift: for instrument-response shift, zero-label CORAL raises the median ratio of performance to deviation (RPD) from 0.61 for direct transfer of the source model to 2.01, reaching a common screening-level threshold ($\text{RPD} \geq 2$ is an empirical, application- and variance-dependent screening convention rather than a guarantee of industrial-grade accuracy[6], and 2.01 is a benchmark-level median: about 53% of the 30 instrument tasks individually reach $\text{RPD} \geq 2$ and 83% reach $\text{RPD} \geq 1.4$); for seasonal, cross-laboratory and cross-origin shift, CORAL raises RPD from 0.84, 0.72 and 0.47 to 1.27, 1.02 and 0.75 respectively, a partial recovery that falls short of that threshold. It should be noted that CORAL is a transductive adaptation: it estimates its second-order moments from the unlabelled target spectra, so “zero labels” means no target labels, not no target data, and the alignment is refitted per target domain rather than deployed inductively to a single new sample.

Organising the most frequent winning method at each label budget under each type of shift into an in-sample frequency map (Fig. 3) shows the budget

Table 4

Oracle upper-envelope comparison: best simple correction vs. best deep/physics representation, paired by task (median across the five fixed seeds, Cliff's δ and the paired Wilcoxon test). Because the per-task best is taken within each family, this is a descriptive upper envelope, not an operational selection rule; a selection-free reanalysis is given in Table 5.

Shift type	n	Tasks	RMSEP _S	RMSEP _D	NRMSEP _S	NRMSEP _D	S-win %	Cliff's δ	Wilcoxon P
Instrument	0	30	0.241	0.359	0.497	0.798	93	0.33	$< 10^{-4}$
Instrument	20	30	0.283	0.332	0.522	0.786	77	0.25	0.002
Instrument	40	30	0.279	0.332	0.510	0.767	77	0.25	0.002
Season	10	12	2.037	2.697	0.853	1.094	92	0.83	0.001
Season	20	12	1.975	2.536	0.833	0.974	92	0.62	0.002
Origin/year	0	72	1.960	1.945	1.336	1.352	58	+0.07	0.076
Origin/year	5	72	1.521	1.613	1.142	1.163	56	+0.20	0.010
Origin/year	20	72	1.292	1.385	0.998	1.059	72	+0.20	$< 10^{-4}$
Lab (MIR)	10	12	11.270	7.954	0.945	0.560	0	-0.26	$< 10^{-3}$

Note: RMSEP is in the units of the target variable of each benchmark ($^{\circ}$ Brix or dry matter % for apple/mango, constituent % for corn, total carbon % for soil); NRMSEP = $\text{RMSEP}/\sigma_{y,\text{tgt}} = 1/\text{RPD}$ is dimensionless (medians over tasks of the task-wise value, script 75); subscripts S and D denote the simple and deep families, and "S-win %" the simple-win percentage. Each family is represented at each budget by the members that can be run at it: "simple" is CORAL at $n = 0$ and the slope/bias correction at $n > 0$; "deep" is the best of the four zero-shot models at $n = 0$ and the best of the fine-tuned CNN and Phys+BL at $n > 0$. The label-free members therefore appear in the $n = 0$ rows, and the budget rows compare what the labels buy in each family, not the label-free result carried forward. "Simple-win %" is the percentage of the paired tasks on which the simple family has the smaller error, the paired complement to Cliff's δ ; a negative δ (win below 50%) indicates that the deep methods are better. All results use the five fixed seeds [20060515, 20041210, 19810915, 2023, 2024].

Table 5

Selection-free comparison of the simple and deep families (no per-task best-of-family selection), computed from the same runs as Table 4. Two variants: a pre-specified representative pair (label-free CORAL vs. its deep counterpart DeepCORAL) and the family mean (mean over all methods in each family per task). A positive Cliff's δ and a win fraction above 0.5 both indicate that the simple family has the smaller error.

Comparison	Shift type	n	RMSEP _S	RMSEP _D	NRMSEP _S	NRMSEP _D	S-win %	Cliff's δ	Wilcoxon P
CORAL vs. DeepCORAL	Instrument	0	0.241	0.639	0.497	0.984	100	+0.48	$< 10^{-8}$
CORAL vs. DeepCORAL	Season	0	1.959	2.738	0.791	1.046	83	+0.57	0.021
CORAL vs. DeepCORAL	Origin/year	0	1.960	2.246	1.336	1.499	74	+0.24	$< 10^{-5}$
CORAL vs. DeepCORAL	Lab (MIR)	0	11.33	9.94	0.991	0.818	25	-0.11	0.34
Family mean	Instrument	0	0.241	0.570	0.497	0.929	100	+0.46	$< 10^{-8}$
Family mean	Instrument	20	0.283	0.426	0.522	0.892	90	+0.41	$< 10^{-3}$
Family mean	Season	20	1.975	2.607	0.833	0.996	92	+0.68	0.001
Family mean	Origin/year	0	1.960	2.195	1.336	1.464	75	+0.24	$< 10^{-6}$
Family mean	Origin/year	20	1.292	1.512	0.998	1.125	92	+0.38	$< 10^{-11}$
Family mean	Lab (MIR)	20	11.08	7.12	0.905	0.521	0	-0.31	$< 10^{-3}$

Note: same units, NRMSEP definition, subscripts and seeds as Table 4. "Simple-win %" is the percentage of paired tasks on which the simple family has the smaller error (the paired complement to Cliff's δ). The representative pair is defined a priori as the two second-order moment alignment methods (CORAL, DeepCORAL) and is compared at zero labels, where both are label-free. The family mean is the per-task unweighted mean over the family members that can be run at that budget (at $n = 0$, CORAL against the four zero-shot deep models; at $n > 0$, the slope/bias correction against the fine-tuned CNN and Phys+BL), so the deep family is summarised over more constituents than the simple family. Cliff's δ is the same unpaired dominance statistic as in Table 4, so the two tables are directly comparable; the paired information is carried by "Simple-win %". Only budgets 0 and 20 are shown; over the full grid (script 86) the direction is unchanged at 5/10/40 on the instrument, seasonal and origin/year benchmarks, with family-mean δ of +0.39 to +0.46 for instrument, +0.24 to +0.45 for origin/year and +0.54 to +0.86 for season; over those three shift types and the whole grid the family-mean δ runs from +0.24 to +0.86 at $P \leq 0.007$. Collapsing each source-target domain pair's tasks to their median (98 pairs instead of 126 tasks; script 101) leaves every direction unchanged and the effect sizes at least as large, with family-mean δ of +0.56 to +0.62 for instrument, +0.54 to +0.86 for season, +0.24 to +0.45 for origin/year and -0.56 to -0.67 for soil at positive budgets; the significance survives for season ($P \leq 0.007$), origin/year ($P < 10^{-6}$) and soil above zero labels ($P = 0.031$), but not on the instrument benchmarks, which collapse to eight pairs ($P = 0.20$ at every positive budget). Cross-laboratory soil is the exception: the two families are not separated there at zero labels ($\delta = -0.08$, $P = 0.47$), and from five labels onwards the deep family has the smaller error on all 12 tasks at every budget ($\delta -0.25$ to -0.32 , $P = 5 \times 10^{-4}$).

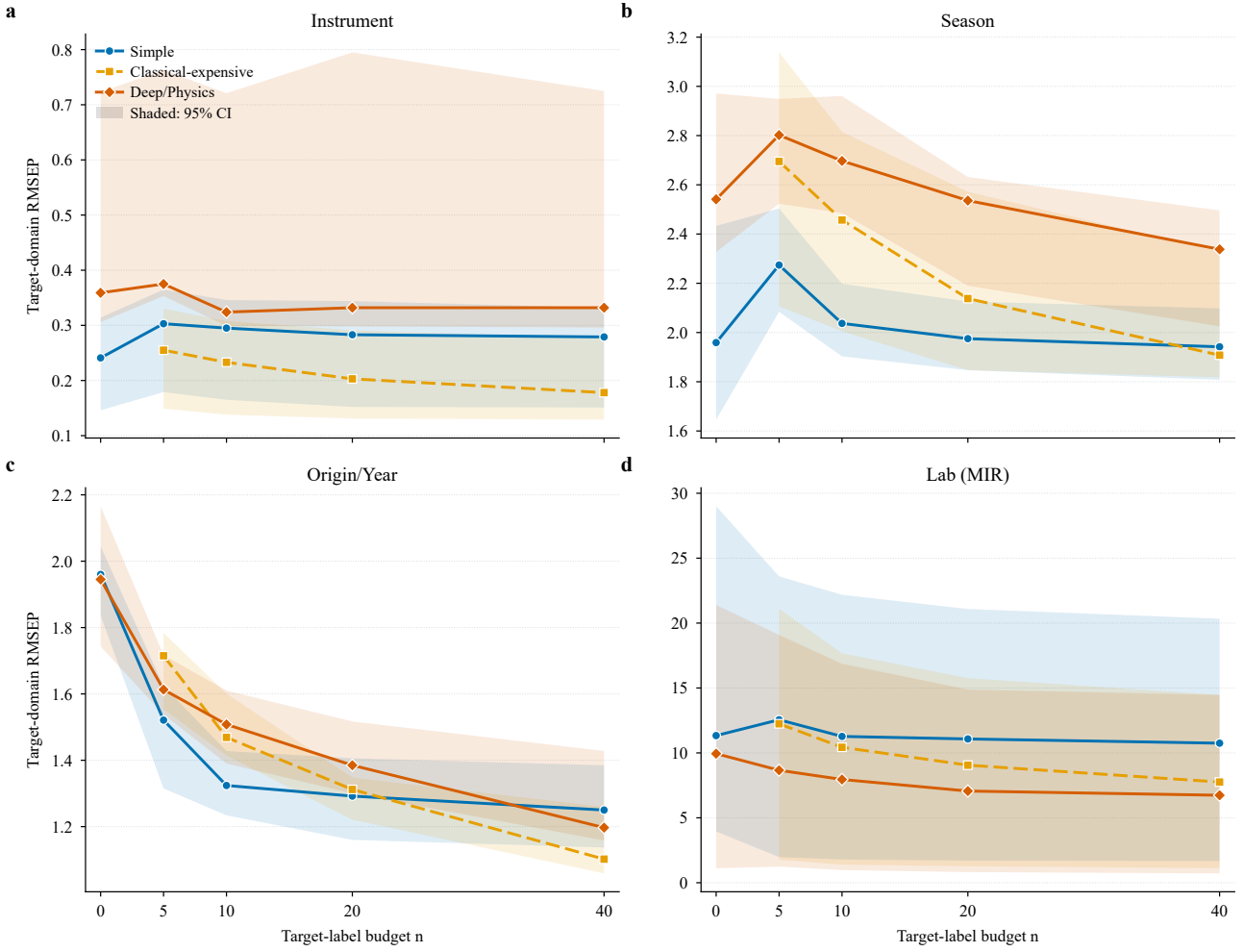


Figure 2: Reckoning curves of the three method families (best simple correction, classical model update, best deep/physics representation) against target-label budget under the four types of domain shift: median target-domain RMSEP with task-clustered 95% confidence bands

Note: the ordinate of each panel is in the native target units of the corresponding benchmark (instrument = corn constituents in % and tablet weight/hardness/assay in their own units; season = mango dry matter %; cross-origin/year = apple °Brix; cross-laboratory = soil total carbon and sand in %); the units differ, so the axes are scaled separately and values are not comparable across panels. The instrument panel pools corn and tablets, whose analytes are in different units, so it is to be read for the shape of each family's curve against budget and not for the absolute level; the cross-benchmark statements rest on the normalised endpoint of Tables 4 and 5 instead. Medians are taken across the five fixed seeds, and the shaded bands are cluster-bootstrap 95% confidence intervals with the task as the clustering unit.

dependence directly. The map is taken over the twelve candidates of Table 3, each at the budgets it consumes, so every row compares what is available at that budget: the $n = 0$ row of Fig. 3a is a comparison among the six label-free candidates (direct transfer and transductive CORAL among the classical ones, together with the four zero-shot deep models), and the rows above it among the six that take target labels. CORAL, which is designed for precisely this setting, is the most frequent winner on the instrument (93%), seasonal (58%) and cross-origin (50%) benchmarks; on cross-laboratory soil it is not, where DeepCORAL wins 42% of the tasks against CORAL's 25%, on the same benchmark that reverses in Table 4. Restricted to the two classical label-free candidates alone, CORAL wins on 75–100% of the tasks of every shift type. Part of that advantage

is one of applicability, because CORAL can run at this budget while most supervised methods cannot, rather than a difference in performance among all methods under the same conditions, so the map is a descriptive, conditional in-sample summary and not a method-selection rule. As target labels accumulate, instrument shift moves to model updating and seasonal and origin/year shift to the two-parameter slope/bias correction, with origin/year passing to a target-only model at 40 labels. Cross-laboratory soil moves instead to the fine-tuned CNN and stays there at every budget from five labels to forty (50–75% of its twelve tasks); it is the only benchmark on which a deep method is ever the most frequent winner, at any budget. Read as a heuristic, the map suggests trying the simplest correction that matches the budget before a deep

model, except on the one benchmark where the map itself points at a deep model; but because it is fitted on these same five benchmarks, we tested rather than assumed that it carries over.

It does not, whereas a fixed prescription does (Table 6). With each benchmark held out in turn and the map re-derived from the other four over the same twelve candidates, so that the only difference from Fig. 3a is that the benchmark being predicted did not contribute to the map, using it as a look-up rule has a pooled median regret of 11.2% relative to the per-task oracle when the shift type is unknown, and fails outright when the shift type is known and the rule is carried between the two instrument benchmarks from five labels onwards (regret 86–104%, pooled 80.3%, because the tablet benchmark selects a fine-tuned CNN that corn does not want). The simplest applicable correction (CORAL at zero labels, SBC otherwise) needs no knowledge of the shift type and had the lowest median leave-one-benchmark-out regret of the prescriptions tested, 2.8% (95% CI 1.7–4.8%), landing within 10% of the oracle on 67% of the 630 task–budget cells; its weakest budget is 40 labels (6.8%), where model updating (6.9%) and target-only calibration (8.8%) are no better. Always updating the model costs 9.9%, always calibrating on the target alone 26%, and defaulting to a deep model 20.8% (18.5–24.3%) with only 30% of cells within 10% of the oracle. This ordering does not depend on the candidate-set convention, but the size of the frequency map’s failure does. Keeping the label-free candidates in the set at every budget, so that a practitioner with five labels may still decline to spend them, raises every prescription’s regret without changing their order (simplest applicable correction 6.1%, always model update 12.4%, default deep model 24.6%, always target-only 29.8%, frequency map with the shift type unknown 14.5%), but the shift-type-known variant then costs nothing at all (0.0%), because under that convention the map derived on one instrument benchmark selects CORAL rather than a fine-tuned CNN. The outright failure of the shift-type-known rule reported above is therefore specific to the convention that a budget’s candidates are the methods consuming its labels, whereas the failure of the shift-type-unknown rule, which worsens from 9.1% at five labels to 29.4% at forty under the carried-forward convention, is not. The operational content of the frequency map is therefore not the map itself but the prescription it points to, and that prescription requires no diagnosis of the shift.

3.3. Absence of a scaling law from shift magnitude to the crossover budget

The patterns above naturally raise a question: can the crossover label budget n^* at which a simple method catches up with or overtakes a complex one be predicted in advance from measurable quantities of domain shift, thereby forming a “transferability scaling law”? We

measured n^* for each transfer task and regressed it in log–log space on the magnitude of domain shift (maximum mean discrepancy MMD, sliced Wasserstein distance), the intrinsic dimensionality of the target domain and the signal-to-noise ratio; extrapolability was then tested by leave-one-benchmark-out extrapolation (fit on the remaining benchmarks, predict the held-out benchmark).

The answer is negative: the log–log fits of n^* on the above continuous measures of shift have coefficients of determination R^2 of only 0.001–0.09 under the definition used in the main text (at most 0.13 under the n_{usable} definition, and then on only 28 tasks for which it is defined), leave-one-benchmark-out extrapolation reaches a mean absolute error of 0.6–2.2 labels on the original scale for that definition and 3.9–20 labels for the source-versus-target definition, and the coverage of conformal 90% intervals is erratic, between 0.00 and 1.00. The scatter of n^* against MMD (Fig. 4b) is nearly flat, illustrating this negative result directly. Even with the cross-modality mid-infrared soil benchmark included and five fixed seeds used throughout, this conclusion, that no extrapolable scaling relation is found, remains robust. A possible explanation is that domain distances across different types of shift (instrument response, population, season, laboratory) are nearly incommensurable under the metrics used here, the same numerical MMD corresponds to widely different transfer difficulties under different mechanisms, so marginal distribution distances alone can hardly characterise n^* across mechanisms. The scope and strength of this negative result must be delimited carefully. First, it is confined to the family of metrics examined here (MMD, sliced Wasserstein, intrinsic dimensionality) and to the log–log functional form, and does not exclude the existence of a scaling relation under other metrics or representations (for instance in a self-supervised pretrained latent space). Second, the number of independent benchmarks is only $n = 5$ and each type of shift is carried mainly by a single dataset (see Section 3.1 and the collinearity limitation in the Discussion), while n^* is itself a discrete and possibly censored quantity on the budget grid; this conclusion is therefore the empirical statement that no unified scaling law was observed under the present design, not a proof that none exists. What carries over is not the ranking of methods over discrete types of shift either, which Section 3.2 shows fails as a rule on an unseen benchmark, but the budget-indexed prescription of Table 6; neither of them is a quantitative extrapolation from continuous shift magnitude.

3.4. Structure of the shift as the quantity that tracks where simple corrections win

Section 3.3 showed that the magnitude of the shift does not predict when a simple correction suffices. We

Table 6

Prospective (leave-one-benchmark-out) regret of using the frequency map as a rule and of fixed prescriptions, relative to the per-task oracle over the twelve candidates of Table 3, each at the budgets it consumes: median regret (%) by label budget, pooled median with its 95% task-clustered bootstrap interval, and the fraction of task–budget cells within 10% of the oracle (script 73).

Prescription	$n = 0$	$n = 5$	$n = 10$	$n = 20$	$n = 40$	Pooled [95% CI]	Within 10%
Simplest applicable correction (CORAL at $n = 0$, SBC otherwise)	0.0	5.6	1.1	2.7	6.8	2.8 [1.7, 4.8]	67%
Always model update		12.6	12.4	8.3	6.9	9.9 [7.0, 12.6]	50%
Always target-only calibration		43.4	32.4	17.1	8.8	26.3 [22.4, 29.3]	24%
Frequency map as a rule, shift type unknown	0.0	18.2	15.5	12.9	13.8	11.2 [9.2, 13.8]	48%
Frequency map as a rule, shift type known (instrument tasks only)	0.0	86.2	97.9	90.6	103.7	80.3 [66.8, 93.9]	31%
Default deep model (zero-shot CNN at $n = 0$, fine-tuned CNN otherwise)	26.9	21.0	26.0	21.5	14.8	20.8 [18.5, 24.3]	30%

Note: regret = $\text{RMSEP}_{\text{chosen}}/\text{RMSEP}_{\text{oracle}} - 1$ on the same test samples; entries are medians over the 126 tasks (over the 30 instrument tasks for the shift-type-known rule, which can only be carried between corn and tablet). “Pooled” pools all budgets at which the prescription is defined (630 cells, or 504 without $n = 0$). The frequency map is re-derived for each held-out benchmark from the other four over the twelve candidates of Table 3, each at the budgets it consumes, with each task weighted equally; weighting each training benchmark equally instead, so that apple’s 72 tasks do not decide the rule on their own, gives 11.8% [10.0, 14.3]. The oracle at $n = 0$ is taken over the label-free candidates only.

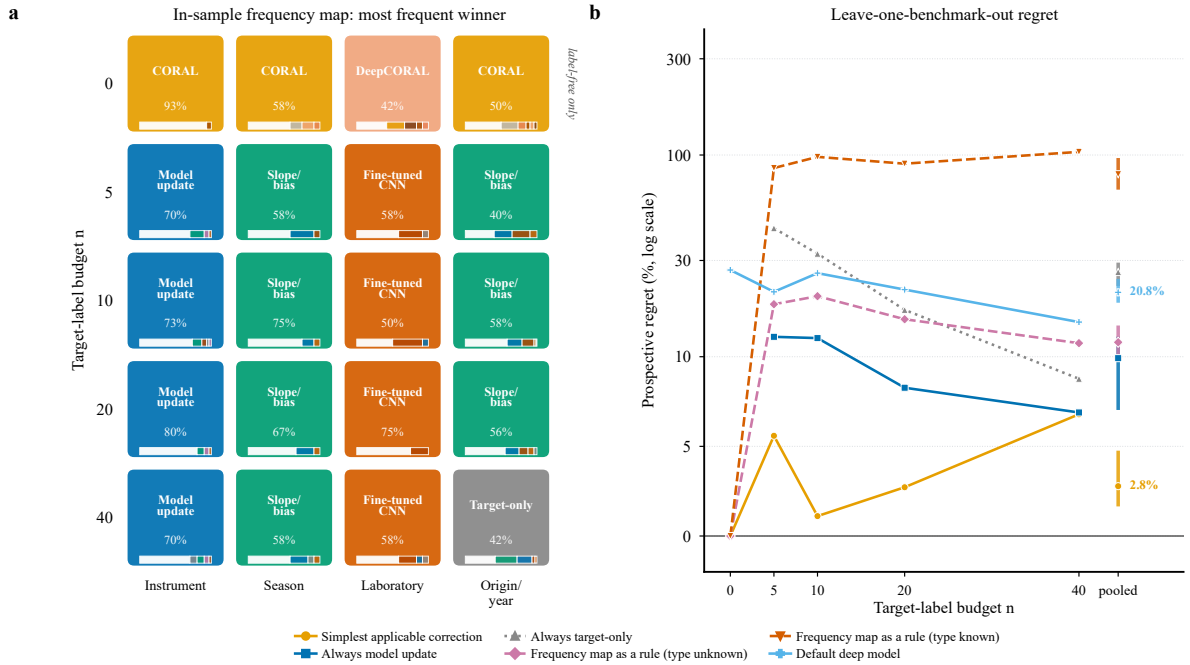


Figure 3: What wins in sample, and what that costs on a held-out benchmark. (a) In-sample frequency map over label budget (rows) and shift type (columns) across the twelve candidates, each at the budgets it consumes: each cell names the most frequent winning method and its share among tasks of that type, and the thin bar inside the cell gives the full win-frequency composition of that cell. At $n = 0$ only the six label-free candidates apply (direct transfer, CORAL and the four zero-shot deep models); above it, only the six that take target labels. (b) Leave-one-benchmark-out regret of six prescriptions relative to the per-task oracle over the twelve candidates of Table 3, each at the budgets it consumes, by label budget and pooled with its 95% task-clustered bootstrap interval: the frequency map used as a rule does not carry over to an unseen benchmark, whereas the simplest applicable correction does

Note: the candidates in (a) are the twelve of Table 3, each entered at the budgets it consumes; the winner is the method attaining the smallest target-domain RMSEP on the largest number of tasks of that (shift type, label budget) combination, so the percentage is a share of the whole field and a cell can be won with well under half of it. The $n = 0$ row is a comparison among the six label-free candidates only, because the other six need target labels. Cell counts are as small as 12 for season and laboratory, so a single task moves a soil or mango cell by eight percentage points. In (b) regret = $\text{RMSEP}_{\text{chosen}}/\text{RMSEP}_{\text{oracle}} - 1$ on the same test samples, the ordinate is on a symmetric-log scale, zero is the oracle, and the numerical values with their intervals are given in Table 6.

therefore asked whether its structure does, using the two label-free diagnostics of Section 2.4 (Eqs. (1)–(3)) on all 98 source–target domain pairs (126 tasks) of the five benchmarks (Table 7, Fig. 4). Decomposing the source–target distance of each pair into the part removed by matching the means, the further part removed by matching the second moments (CORAL) and the higher-order residual that neither can touch gives a clear ordering. On the two instrument benchmarks 92% of the sliced-Wasserstein distance is removed by the first two moments (87–90% by the mean alone) and the residual is 8%; on the laboratory, seasonal and origin/year benchmarks the low-order share falls to 73–78% and the residual rises to 22–27% (with the MMD² distance the ordering is the same, 99–99.9% against 90–96%). Instrument shift, in other words, is almost entirely a low-order transformation of feature space, whereas the population-type shifts carry a substantial component that no first- and second-moment correction can remove.

This structural share, and not the distance itself, is what tracks the outcomes. Across the 126 tasks the raw MMD² is unrelated to how much of the zero-label error CORAL recovers (Spearman $\rho = 0.10$, 95% CI $[-0.12, 0.31]$ from a bootstrap clustered on domain pair) and to the deep-minus-simple margin at 20 labels ($\rho = 0.08$ $[-0.19, 0.32]$), although it is weakly related to how badly direct transfer fails ($\rho = 0.27$ $[0.11, 0.42]$) and to the margin at zero labels ($\rho = 0.26$ $[0.04, 0.45]$). The low-order share f_{CORAL} , by contrast, correlates with the CORAL gain ($\rho = 0.53$ $[0.30, 0.66]$), negatively with the error that remains after alignment ($\rho = -0.41$ $[-0.61, -0.11]$), and positively with the margin by which the simple family beats the deep family ($\rho = 0.49$ $[0.32, 0.61]$ at zero labels, 0.32 $[0.07, 0.49]$ at 20 labels); the sliced-Wasserstein version gives the same picture ($\rho = 0.49$, -0.47 , 0.44 and 0.24). These three correlations do not carry equal weight. f_{CORAL} and the CORAL gain both measure how well a second-moment alignment fits a given domain pair, one in distribution space and one in error space, so their agreement is closer to a consistency check than to an independent finding; the claim that the structure of a shift explains where a simple correction suffices rests on the third correlation, the one with the margin between the two families, which is the weakest of the three but whose interval excludes zero at both budgets. Held out one benchmark at a time it keeps its sign and excludes zero in all five cases at zero labels, and in four of the five at 20, so the zero-label statement carries across benchmarks and the 20-label one does not (Supplementary Section S15). That correlation has a concrete reading at the benchmark level, and a limit. The two instrument benchmarks, whose low-order share is the highest (92%), are the ones on which the deep family loses most heavily, and cross-laboratory soil is the one on which it wins

(Table 7, $\Delta_{20} = -0.36$). The share alone, however, does not single soil out among the population-type benchmarks: seasonal mango (73%) and cross-origin apple (74%) leave a larger residual than soil (78%) and the simple family wins on both. What distinguishes soil is what that residual costs in error space: a label-free alignment recovers only 11% of its zero-label error, against 21–23% on mango and apple and 71–77% on the instrument benchmarks, and leaves it at NRMSEP 0.99, the level of predicting the target mean. The low-order share therefore identifies the class of shifts in which a two-parameter correction cannot reach the residual, not which member of that class a learned representation will win. The relation carries across benchmarks: holding out each benchmark in turn, the correlation between f_{CORAL} and the CORAL gain on the remaining benchmarks stays between 0.29 and 0.68 with every lower confidence limit above zero, and it keeps its sign within corn ($\rho = 0.40$, 24 tasks), mango (0.55, 12) and apple (0.23, 72). The one exception is the soil benchmark, the only mid-infrared benchmark here, within which the relation reverses ($\rho = -0.76$ over its 12 tasks from six pairs); with $n = 5$ benchmarks we report this as a limitation rather than explain it away, and it bounds the diagnostic to the modality and benchmarks tested.

The second diagnostic answers a different question. The low-rank residual amplification A correlates with the zero-label error of direct transfer ($\rho = 0.49$ $[0.34, 0.62]$) and with the CORAL gain ($\rho = 0.58$ $[0.42, 0.70]$), with the same sign in every held-out benchmark (leave-one-benchmark-out ρ 0.56–0.63 against the gain and 0.32–0.58 against the zero-label error) and within four of the five benchmarks (within-benchmark ρ 0.27–0.72; tablet, with six tasks from two pairs, cannot be estimated); the held-out variant gives nearly the same values ($\rho = 0.54$ and 0.48). The diagnostic threshold pre-registered for the apple case study, $\rho \geq 0.5$, was not reached within that case study (Supplementary Section S8) and is not reached here either: on the pooled five-benchmark evidence A falls 0.008 short of it. We state the shortfall rather than round it away, because it is the difference between a warning indicator and a validated decision rule, and A is only the former. But A is unrelated to the error that remains after alignment ($\rho = -0.09$ $[-0.29, 0.16]$) and to the deep-minus-simple margin at 20 labels ($\rho = 0.16$ $[-0.07, 0.34]$), while at zero labels it does track that margin ($\rho = 0.42$ $[0.24, 0.57]$). A is therefore a label-free warning, computable before any target label is acquired, of whether direct transfer is safe and of how much a label-free alignment will recover; it says nothing about the difficulty that is left once the low-order part of the shift has been removed, which is where a deep representation would have to earn its cost. Together the two diagnostics turn the descriptive pattern of Sections 3.1–3.3 into a testable statement:

Table 7

Label-free structure of the shift on the five benchmarks (script 74): share of the source–target distance removed by matching the first moment (f_{mean}) and the first two moments (f_{CORAL}), the higher-order residual $1 - f_{\text{CORAL}}$, and the low-rank residual amplification A , alongside the benchmark outcomes they are compared with.

Benchmark	Shift type	Pairs / tasks	f_{mean}	f_{CORAL}	Residual	f_{CORAL} (MMD ²)	A	CORAL gain	Δ_0	Δ_{20}
Corn	Instrument	6 / 24	0.90	0.92	0.08	0.999	7.42	0.71	+0.50	+0.45
Tablet	Instrument	2 / 6	0.87	0.92	0.08	0.992	3.22	0.77	+0.17	+0.01
Soil (OSSSL)	Laboratory	6 / 12	0.56	0.78	0.22	0.964	2.43	0.11	−0.18	−0.36
Mango	Season	12 / 12	0.54	0.73	0.27	0.905	2.74	0.21	+0.19	+0.24
Apple	Origin/year	72 / 72	0.64	0.74	0.26	0.953	6.09	0.23	+0.18	+0.18

Note: shares are computed with the sliced Wasserstein-1 distance unless stated (the MMD² column gives the corresponding f_{CORAL}); each is a property of the ordered domain pair, evaluated on SNV spectra without labels, and benchmark values are medians over tasks. CORAL gain = $1 - \text{NRMSEP}_{\text{CORAL}}/\text{NRMSEP}_{\text{direct}}$ at zero labels; Δ_n is the family-mean deep-minus-simple NRMSEP at n labels, positive favouring the simple family, taken as the median over the tasks of the benchmark; the benchmark-level test of Section 3.1 averages the same per-task difference instead, which changes the sign on tablets at $n = 20$ (+0.01 as a median, −0.03 as a mean) and on no other benchmark. Correlations across the 126 tasks are given in the text.

simple corrections win where the shift is low-order, and how low-order a shift is can be read from unlabelled spectra.

3.5. The apple case study in brief

The in-house apple data are the largest single contributor to the benchmark, supplying 72 of the 126 transfer tasks and the whole of the origin/year shift class; without them the design would cover three types of shift rather than four, and no benchmark would combine a change of instrument generation with a change of population. They were additionally examined as a case study of the physics-informed architecture on their own pipeline, for two reasons on which the main line depends: the label-free index A that Section 3.4 generalises to all five benchmarks was found here, and this is the only setting in which a representation built around the physics of absorbance can be tested against a two-parameter correction funded by the same target labels. The complete design, tables and figures are given in Supplementary Sections S1–S12, and only what bears on the main line is summarised here. The case study contrasts the 2018 (instrument A) and 2025 (instrument B) seasons over 602 source→target scenarios formed from single and multiple origins (24 same-year cross-origin, 578 cross-year), with its own preprocessing (wavelet denoising, Savitzky–Golay smoothing and band-wise standardisation; scatter correction deliberately omitted) and its own label protocol (the deep models use 20% of the target labels for early stopping only, the chemometric baselines none), so its absolute errors are not comparable with those of the unified benchmark (Supplementary Sections S2–S3). Three findings carry over.

First, the additive low-rank structure that holds within an origin destabilises across origins: the residual amplification A of Section 2.4 ranged from 1.08 (Shaanxi→Shandong, the geographically closest pair) to 4.55 (Xinjiang→Shaanxi) over the 12 same-year ordered origin pairs, all significantly above 1, and correlated with cross-domain transfer risk (median

Spearman $\rho = 0.31$; Supplementary Section S8). It is this label-free warning that Section 3.4 generalises to the five benchmarks.

Second, in the uncorrected setting every method fails in absolute terms (RPD below 1) and Phys+BL is the least bad: mean RMSE 1.892 °Brix over the 572 scenarios in which every method converged, 7.7% below the CNN given the same early-stopping labels (2.050 °Brix; cluster-robust 95% CI excluding zero under scenario and target-set clustering, and touching zero under the most conservative source-set clustering). The ordering is unchanged under a strict zero-label protocol in which the deep models are early-stopped on a source hold-out (Phys+BL median 1.77 °Brix, the lowest of the seven methods compared there), at 0% and 15% outlier removal, and across scenario strata (Supplementary Section S9). A five-seed ablation attributes the advantage to the low-rank additive factorisation architecture rather than to the Beer–Lambert reconstruction loss: removing the loss leaves accuracy essentially unchanged, adding it to a generic CNN changes nothing the test can resolve, and its real effect is to cut the divergence rate of the physics model roughly ninefold (3.6% to 0.4% of scenario–seed runs; Supplementary Section S10).

Third, once the same 20% of target labels are spent instead on a two-parameter slope/bias correction, the classical baselines converge to 1.58–1.61 °Brix (PLSR 3.33→1.58, SVR 2.56→1.61 °Brix) and overtake the corrected physics model (1.89→1.84 °Brix); the misclassification rate at the 13 °Brix grade boundary stays at 34.1–36.6% and the RPD at 1.02 or below for every corrected method, so the most economical option on these data is not a production-ready one (Supplementary Section S11). The case study thus adds two things to the main line and takes nothing from it: a label-free indicator of shift structure that turned out to generalise, and a direct demonstration that a physics-informed representation’s advantage over a generic network does not become an advantage over

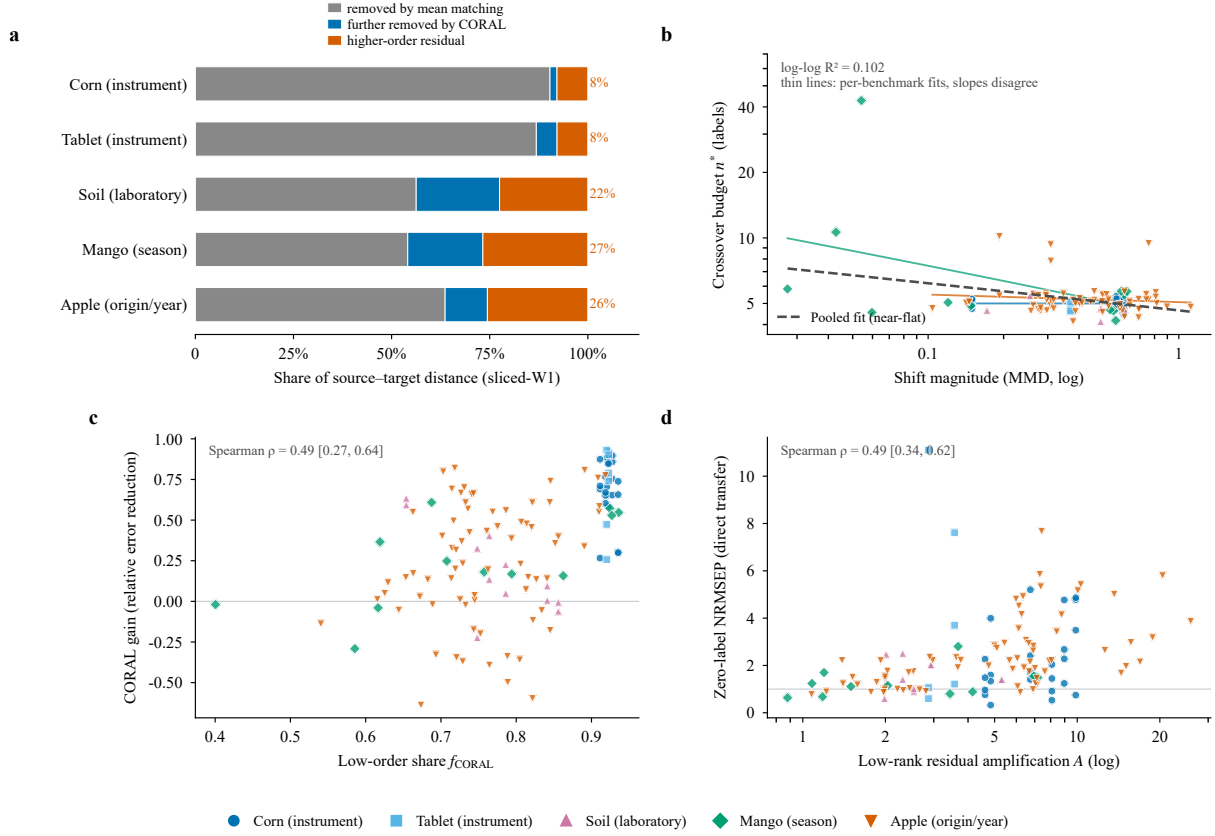


Figure 4: Structure, not magnitude, of the shift. (a) Decomposition of the source–target sliced-Wasserstein distance of each benchmark into the share removed by matching the mean, the further share removed by second-order (CORAL) alignment and the higher-order residual. (b) Crossover budget n^* against shift magnitude (MMD), with per-benchmark fits whose slopes disagree and a near-flat pooled fit. (c) Low-order share f_{CORAL} against the relative error reduction achieved by label-free CORAL, one point per task. (d) Low-rank residual amplification A against the zero-label NRMSEP of direct transfer

Note: in (a) bars are benchmark medians over domain pairs; in (b) points are the tasks for which n^* is defined, jittered vertically because n^* takes only the four grid values, and both axes are logarithmic; in (c) and (d) points are the 126 tasks, coloured by benchmark, with Spearman ρ and its 95% domain-pair-clustered bootstrap interval, and A is on a logarithmic abscissa. Raw shift magnitude (MMD^2) is not associated with the CORAL gain ($\rho = 0.10$, see text), which is the contrast between (b) and (c).

a two-parameter correction once the same labels are spent on the latter.

4. Discussion

4.1. Low-order shift structure as the reason simple corrections win

In a unified head-to-head comparison over 126 tasks from five benchmarks, the more expensive deep and physics-informed representations were not found to be better than the simplest classical corrections at any label budget on four of the five benchmarks, and the size of the simple family’s advantage was associated with the benchmark and its shift type rather than with the continuous magnitude of the shift (Tables 4–5). The fifth, cross-laboratory soil in the mid-infrared, went the other way at every positive budget, and it is the benchmark the same explanation predicts would. One explanation, that inter-instrument differences and whole-season displacements are largely low-order (first-

and second-moment) transformations of feature space, which is exactly the structure that CORAL and slope/bias correction target, is supported directly by the structure analysis of Section 3.4: 92% of the source–target distance on the instrument benchmarks is removed by matching the first two moments, against 73–78% on the seasonal, laboratory and origin/year benchmarks, and this low-order share, not the distance itself, correlates with how much a label-free alignment recovers ($\rho = 0.53$) and with the margin by which the simple family beats the deep family ($\rho = 0.49$). In the taxonomy of Section 2.1, instrument shift is a covariate shift that a low-order map of $\mathbf{x}$ can undo, whereas the population shifts are prior and conditional shifts whose higher-order residual (22–27%) no first- and second-moment correction can remove; that residual is where target labels become necessary in principle. On two of the three population-type benchmarks the deep methods still failed to convert their capacity into accuracy under the matched budgets of this study,

but on cross-laboratory soil they did. That is the benchmark on which a label-free alignment recovers the least of the zero-label error and leaves it at the level of predicting the target mean: from five target labels onwards the deep family is better on all 12 tasks, and its NRMSEP falls to 0.50 while the simple family stops at 0.87. The reading is therefore not that a learned representation never pays, but that it pays where the shift leaves a residual that no low-order correction can reach. A deep network trained on source labels has discriminative features that are not naturally invariant to low-order shifts of the target domain, and with at most 40 target labels it has no mechanism to learn the residual either, whereas the two-parameter correction, being low-dimensional, has a small estimation variance and is naturally favoured at small budgets. The practical implication is that, under the protocol and budgets evaluated here, deploying deep domain adaptation by default lacks empirical support, and method selection is better started from the simplest applicable correction.

4.2. Deployment when the type of shift is unknown

A recommendation that begins with “determine the type of shift” is of limited use, because in deployment the shift is often unknown, mixed or uncontrolled. The leave-one-benchmark-out test of Section 3.2 makes the recommendation independent of that diagnosis. Used as a look-up rule, the frequency map does not transfer to an unseen benchmark, and its shift-type-known variant fails outright between the two instrument benchmarks; the prescription that survives is the one that needs no diagnosis at all, the simplest applicable correction, CORAL when no target label is available and a slope/bias correction otherwise, with a median regret of 2.8% relative to the per-task oracle over the twelve candidates of Table 3, each at the budgets it consumes, and 67% of task-budget cells within 10% of it, against 20.8% and 30% for defaulting to a deep model (Table 6). Its weakest point is the 40-label budget, where model updating has caught up; across the grid it stays within 7% of the oracle (5.6%, 1.1%, 2.7% and 6.8% at 5, 10, 20 and 40 labels). Its advantage over the runner-up is not decisive, and because all prescriptions are scored on the same cells this can be tested pairwise: on the budgets where both are defined ($n \geq 5$), the median paired regret difference between the simplest applicable correction and always updating the model is -3.6 percentage points with a 95% task-clustered interval of $[-9.7, +0.5]$, and against the frequency map used as a rule -0.8 $[-5.3, 0.0]$; only against a default deep model (-15.2 $[-20.2, -10.9]$) and against calibrating on the target alone (-16.0 $[-20.2, -14.0]$) is the difference separated from zero (script 87). The prescription is therefore the cheapest member of a group of prescriptions that this evidence

cannot fully separate, not a uniquely optimal rule; what the evidence does separate is that group from defaulting to a deep model. Two label-free diagnostics can be computed before any label is bought. The low-rank residual amplification A says whether direct transfer is safe and how much a label-free alignment will recover, and the residual share $1 - f_{\text{CORAL}}$ says how much of the shift a low-order correction will leave behind. On the two benchmarks where a large residual share coincides with an error still near the target standard deviation after alignment (soil at NRMSEP 0.99 and apple at 1.34 after CORAL), no label-free method reached a usable level; on soil, however, a fine-tuned deep model does, reaching NRMSEP 0.50 once 20–40 target labels are available, which is the concrete form the residual takes: it has to be bought, and on that benchmark a learned representation buys it more cheaply than a two-parameter correction. Whether the residual share by itself identifies such cases in advance is not established by five benchmarks, and reading the error itself requires reference labels, so the pairing is an observation about these two benchmarks rather than a validated failure test. What this study does not provide is a method that overcomes such shifts. Its contribution is the measured cost of the available prescriptions and the diagnostics that tell a user, from spectra alone, which regime they are in, in the tradition of the comparative evaluations of calibration transfer [11, 45] rather than of new transfer algorithms.

4.3. Shift structure rather than shift magnitude as the informative quantity

One natural expectation is that, since shift is measurable, the crossover budget n^* should be predictable from distribution distances or intrinsic dimensionality, forming a “transferability scaling law”. Tested directly on the 126 tasks, the answer is negative within the present design: log-log fits of n^* on shift magnitude reach R^2 of only 0.001–0.13, leave-one-benchmark-out extrapolation has a mean absolute error of 0.6–20 labels, and conformal coverage is erratic. Section 3.4 identifies why: the same numerical MMD corresponds to shifts of very different composition, and it is the composition, the share removable by low-order corrections, that tracks what the corrections achieve. This is not a scaling law either: the structural share explains the direction and size of the simple family’s advantage (ρ 0.3–0.5), not the exact budget at which the families cross, and it reverses within the soil benchmark, the same benchmark on which the between-family direction also reverses; the two reversals are therefore one fact seen twice rather than two independent failures. The methodological lesson is that predicting transfer difficulty from marginal distribution distances does not work in this setting, whereas decomposing the shift into what a given correction can and cannot remove does carry information across benchmarks.

4.4. Boundaries of identifiability and external validity

Ten limitations bound how far these conclusions generalise. First, shift mechanism and dataset are nearly collinear: seasonal shift is carried only by mango, cross-laboratory shift only by OSSSL soil and cross-origin/year shift only by our apples, so the number of independent benchmarks is $n = 5$ and the 126 tasks are correlated within benchmarks; mechanism-level statements are an exploratory map rather than a main effect of mechanism, and a strict test needs at least two independent benchmarks per mechanism. The same collinearity limits the structure analysis: the low-order share is a property of the domain pair, so the 126 task-level correlations rest on 98 pairs (hence the pair-clustered intervals) and on five benchmark-level contrasts, and the within-soil reversal shows that the relation is not guaranteed inside a benchmark. Second, the oracle envelope of Table 4 carries a winner's-curse selection compounded by unequal family sizes; the selection-free comparison (Table 5) preserves the pattern and is the primary basis. Third, label-free alignment is transductive: CORAL uses the second-order moments of unlabelled target spectra, and its zero-label usability is not inductive deployment on single new samples; the diagnostics of Section 2.4 likewise need a batch of unlabelled target spectra. Fourth, the tuning budget of the deep baselines is limited, so on small tasks their disadvantage may stem partly from data starvation; "simple wins" holds under the matched compute and tuning budget of this study, and the oracle of Table 6 is bounded by the same budget. Fifth, the leave-one-benchmark-out regret of Table 6 is measured on a benchmark held out from the same five, not on data acquired after the rule was fixed, and the 2.8% is therefore conditional on this set of benchmarks rather than an externally validated deployment cost; a genuinely prospective test on a sixth benchmark remains to be done. Sixth, the frequent zero-label wins of CORAL are partly wins of applicability: methods that need target labels cannot run at that budget (Table 3), so the comparison there is among label-free methods only and must not be read as CORAL being superior under all conditions. PDS is the sharpest case: it consumes no reference value, and under an accounting that charged only reference measurements it would compete in the zero-label row of the two paired-instrument benchmarks, which is exactly the setting it was designed for; the sample-based budget of Section 2.3 keeps it out of that row. Seventh, the potential advantage of complex models may appear at larger label budgets, under more strongly non-linear shifts or with stronger structural priors; the budget grid of this study stops at 40 labels and each shift type is carried mainly by one dataset, so these conditions were not covered systematically. Eighth, the deep family is implemented with the generic

recipe of each method's original description under one fixed optimisation budget, not tuned per benchmark. Taking the best deep method per task, its NRMSEP is below 0.9 on 36% of task-budget cells and below 0.8 on 21% (median 0.981), against 43% and 27% (median 0.952) for the simple family, so the two families are close in this summary and separated by shift type rather than overall. Because the unified protocol records only target-domain error, a failure to transfer and a failure to train are observationally equivalent in it; we therefore ran the same models on held-out samples of their own source domains, where their median NRMSEP is 0.79 against 1.00 for a constant predictor and 0.33–0.80 for a PLS regression on the identical splits (Supplementary Section S14). The deep baselines do learn the spectrum-content relation of their source domain, so what the main tables record is a failure to transfer, not a failure to train, except for Phys+BL on corn and for the whole deep family on apple; those stay at or just under the constant predictor even within their own source domain and are bounded as such in Supplementary Section S14. Quadrupling the optimisation budget uniformly across the five benchmarks leaves this reading unchanged: on apple every run of the three budget-independent deep models returns a bit-identical result, so the budget never bound there; on the other four it improves the source-domain fit without improving target-domain error (Supplementary Section S14). What is compared is therefore the three families as specified in Section 2.2 under one fixed budget, not the best achievable deep model. Ninth, the apple data supply 72 of the 126 tasks, so the analysis was repeated on the 54 tasks of the four public benchmarks (Supplementary Section S15). The comparison within each remaining shift type is untouched and the structural correlations are stronger there, but two quantitative readings depend on apple: the held-out regret of the simplest applicable correction is 8.5% rather than 2.8%, at which level always updating the source model is cheaper (1.7%), and the benchmark-balanced zero-label advantage widens to $+0.22 [-0.07, 0.45]$ over four units. Both prescriptions are classical, so the family-level reading holds; which of them is the cheapest to recommend is decided by the apple tasks. Tenth, the main tables report native-unit RMSEP for interpretability alongside NRMSEP, and a fully hierarchical benchmark→pair→task error model is still limited by $n = 5$, so benchmark-level inference is directional. PDS is a standardisation designed for paired transfer standards and is informative mainly on the paired-instrument benchmarks; its full-window limit, direct standardisation, is the better of the two there but still above both the slope/bias correction and model updating (Supplementary Section S13), so the classical method that binds the comparison is not a standardisation. None of this alters the pragmatic

conclusion that deep domain adaptation does not warrant default deployment on evidence of this kind, but it requires the headline findings to be stated as bounded empirical findings rather than universal laws.

4.5. Role of the apple case study

The case study (Section 3.5; Supplementary Sections S8–S12) asked whether a physics-informed factorisation architecture outperforms classical calibration transfer under cross-origin shift, and its answer delimits the physical prior. The Beer–Lambert-inspired additive structure is useful as a diagnostic: its cross-origin instability, measured without labels, warned of transfer risk on apple and, generalised as A , warns of zero-label failure on all five benchmarks. As an inductive bias it is narrower than hoped: the architecture is the least bad uncorrected model, its advantage over a generic network comes from the low-rank additive factorisation rather than from the reconstruction loss (whose real effect is to stabilise training), and the advantage disappears once the same 20% of target labels fund a two-parameter slope/bias correction of a PLSR model, after which the best corrected model only just clears RPD 1.0 (1.02) on these data, far short of screening-level accuracy. Where the residual share of the shift is large, as it is on apple (26%), the case study therefore corroborates the main line from the opposite direction: a representation designed around the physics of the spectra did not capture the higher-order part of the shift any better than a two-parameter correction, and the cheapest usable option is PLSR with slope/bias correction, with the explicit caveat that it is not production-ready.

5. Conclusions

Across five calibration-transfer benchmarks, 126 transfer tasks, four types of domain shift and two spectral modalities, the simplest classical corrections were at least as good as the deep and physics-informed baselines that each budget admits (four zero-shot models at zero labels, the fine-tuned convolutional and physics-informed networks above it) at every target-label budget from 0 to 40 on four of the five benchmarks. Label-free second-order moment alignment and the two-parameter slope/bias correction led at every budget under instrument, seasonal and cross-origin shift (selection-free family-mean Cliff's δ 0.24–0.86 over the full budget grid, Wilcoxon $P \leq 0.007$). On cross-origin shift that lead is bounded by the positive control of Supplementary Section S14: there the deep baselines stay at or just under a constant predictor even on held-out samples of their own source domains, and a PLS regression on the same splits reaches only 0.80, its weakest on any benchmark, so the cross-origin rows compare methods that all explain little of the source variance. Cross-laboratory soil in the mid-infrared is the exception and the informative one: from five target

labels onwards the deep family was better on all 12 of its tasks ($P = 5 \times 10^{-4}$), reaching NRMSEP 0.50 where the simple family stopped at 0.87. The pattern is the same on the dimensionless NRMSEP; at the benchmark level the pooled advantage of the simple family is established at zero labels (+0.25, 95% CI [0.01, 0.44]) but not at 20 (+0.08, [−0.15, 0.28]).

For a user who does not know the type of shift, the operational result is a prescription rather than a map: CORAL when no target label is available, which presumes a batch of unlabelled target spectra rather than one-at-a-time prediction (Section 4.4), and a slope/bias correction otherwise has a leave-one-benchmark-out median regret of 2.8% relative to the per-task oracle over the twelve candidates, each at the budgets it consumes, against 11.2% for the frequency map used as a rule and 20.8% for a default deep model. The map describes (and the CORAL advantage in its zero-label row is partly one of applicability, since methods that need labels cannot run there); the prescription is what carries over.

The crossover budget could not be predicted from shift magnitude ($\log\text{-}\log R^2 \leq 0.13$), but the structure of the shift could be read from unlabelled spectra: matching the first two moments removes 92% of the source–target distance under instrument shift and 73–78% under population shifts, and that share, not the distance, correlates with the gain from label-free alignment ($\rho = 0.53$) and with the simple family's margin over the deep family ($\rho = 0.49$), the benchmark on which that alignment recovers the least being the one on which the deep family wins. A low-rank residual amplification index computed from spectra alone tracks zero-label failure on all five benchmarks ($\rho = 0.49$, just short of its pre-registered threshold of 0.5), while saying nothing about the difficulty that remains after alignment.

Each shift mechanism is carried mainly by one dataset ($n = 5$), the held-out test is leave-one-benchmark-out rather than new data, and the deep baselines were run with the generic recipe of each method's original description under one matched, limited optimisation budget rather than tuned per benchmark. The apple case study of a Beer–Lambert-inspired architecture confines the physical prior to a diagnostic role: its accuracy edge over a generic network came from the factorisation architecture, not the physical loss, and vanished once the same target labels funded a slope/bias correction, leaving PLSR with slope/bias correction as the most economical, though not production-ready, option on those data.

Data and code availability

The four public benchmarks (Cargill corn, IDRC 2002 tablet shoot-out, OSSSL global soil spectral library, and the Anderson mango dataset) are available from

their original sources cited in Table 2 and are not re-hosted here. The apple near-infrared spectra are internal data of Xinjiang Agricultural University and have not been publicly released; researchers who wish to reproduce or extend the experiments may request access from the corresponding author on a case-by-case basis, subject to the data-sharing policies of the university. The analysis code is available at <https://github.com/2004lryan/015calibration-transfer-label-economics>.

CRediT authorship contribution statement

Panlin Li: Conceptualization, Methodology, Software, Formal analysis, Investigation, Visualization, Writing – original draft. Yuchang Li: Investigation, Validation, Data curation. Longjie Li: Software, Investigation, Data curation. Ya Liu: Resources, Investigation, Data curation. Yutong Feng: Software, Validation. Muhammad Waqar Akram: Writing – review & editing. Junxian Guo: Supervision, Methodology, Resources, Writing – review & editing. Hua Huang: Conceptualization, Supervision, Project administration, Funding acquisition, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) to assist with English-language polishing (translation and editing) of author-written drafts. The tool was not used to generate or interpret experimental data or to formulate the scientific claims. After using this tool the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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